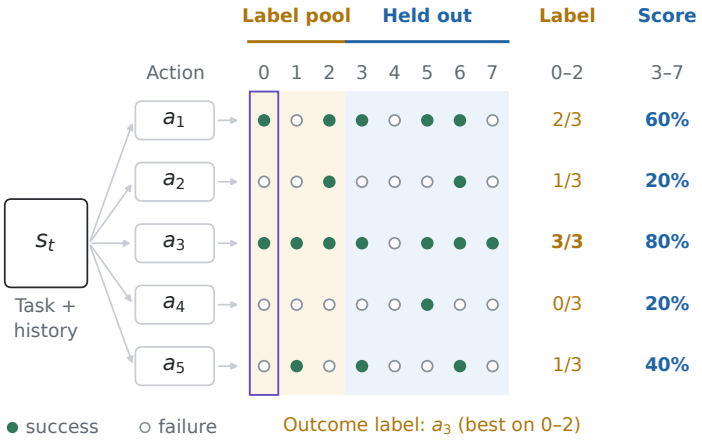
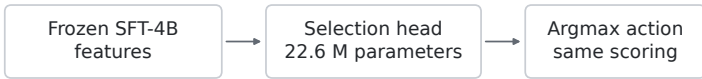
(a) Branch and collect

Each draw: replay s_t → action → frozen actor → judge

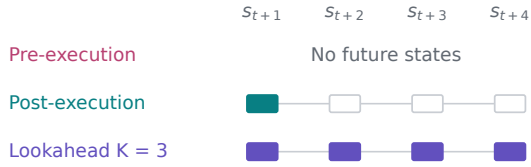


Train a head: 1,380 training states · soft CE



(b) Evidence used for labels

Common input: task, history, s_t, actions



Draw 0 only; K = 1 ends at s_{t+2}.

Final answers are hidden from lookahead.

(c) Score on held-out draws 3–7

Picker	Pick	Score
Uniform	all 5	**44%**
SFT sample	all 5	**44%**
Pre-execution	a_1	**60%**
Post-execution	a_1	**60%**
Lookahead K = 3	a_3	**80%**
Observed outcome	a_3	**80%**
Oracle	a_3	**80%**

Illustrative example · valid draws only. Test mean: 348 states from 86 unseen tasks.